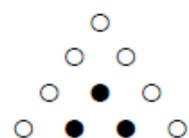
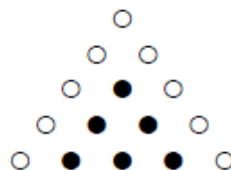


1	YTSS/2025/I/05
	Expand and simplify $a^2 - (4a - b)(3a + b)$. [2]
2	YTSS/2025/I/17
	(a) Factorise completely $x^4 - 16y^4$. [2]
	(b) Hence or otherwise, simplify $\frac{x^4 - 16y^4}{2x^2 - 3xy - 2y^2}$. [2]
3	YTSS/2025/II/08
	The first four terms in a sequence of numbers are given below. $T_1 = 4$ $T_2 = 1 + 9$ $T_3 = 2^2 + 14$ $T_4 = 3^2 + 19$ (a) Find T_5. [1] (b) Without finding the general term, explain why the value of T_n must be even for all values of n. [1] (c) Show that the nth term of the sequence, T_n, is given by $n^2 + 3n$. [2] (d) Find and simplify an expression, in terms of n, for the difference between two consecutive terms, $T_{n+1} - T_n$. [2] (e) Explain why two consecutive terms of the sequence cannot have a difference of 5. [1]
4	SCGS/2025/I/06
	Express $\frac{6a^2}{2a^2 - 8} - \frac{3a}{a + 2}$ as a single fraction in its simplest form. [3]
5	SCGS/2025/I/08
	(a) Expand and simplify $(p - 5)(2p^2 - p)$. [2]
	(b) Simplify $\frac{10 - 13x - 3x^2}{3ax + 3bx - 2a - 2b}$. [3]

6	SCGS/2025/II/08
	(a) The first four diagrams in a sequence are shown below. The diagrams are made from white dots and black dots.  Diagram 1  Diagram 2  Diagram 3  Diagram 4 (i) Write down the number of white dots and black dots in Diagram 5. [2] (ii) Write down an expression, in terms of k, for the number of white dots in Diagram k. [1] (iii) (a) The expression for the number of black dots in Diagram k is $\frac{1}{2}k(k-1)$. Find and simplify an expression, in terms of k, for the total number of dots in Diagram k. [2] (b) Hence, find the diagram with a total of 351 dots. [2] (b) Another sequence is such that the sum of its first n terms is given by $S_n = 3n^2 + 8n$. (i) Show that the nth term of the sequence is given by $T_n = 6n + 5$. [3] (ii) The term $T_p = 125$. Find the value of T_{p-5}. [2]
7	CCHY/2025/I/04
	(a) Simplify $(4n)^2 - (4n+a)(4n-a)$. [1] (b) Hence, without the use of a calculator, evaluate $75^2 - (78)(72)$. [2]
8	SST/2025/I/04(b)
	(b) (i) Factorise completely $4x^2 + 5x - 6$. [1] (ii) Hence, factorise completely $4(2t+2)^2 + 5(2t+2) - 6$. Write your answer as simply as possible. [2]
9	SST/2025/I/06
	Express as a single fraction in its simplest form $\frac{4(x-2)}{x^2-4} - \frac{2(3x-1)}{3x^2+5x-2}$ [3]

10	SST/2025/I/07
	In the sequence, the difference between any two consecutive terms is the same number. $17 \quad x \quad y \quad z \quad 9 \quad \dots$ (a) Find the values of x, y and z. [2] (b) Write down an expression for the nth term of the sequence. [1] (c) Explain why -317 is a term of this sequence. [1]
11	GMSS/2025/II/01
	(a) Simplify $\frac{9-25y^2}{5xy-10y-3x+6}$. [3] (b) Write as a single fraction in its simplest form $\frac{3}{2-x} - \frac{1}{x-2} + 5$. [3] (c) A car which costs $\\$x$ depreciates in value by 3 % every month. Find the value of the car, in terms of x, at the end of 2 years. [2]

Solution

1	$a^2 - (4a - b)(3a + b)$ $= a^2 - (12a^2 + 4ab - 3ab - b^2) \text{ --- [M1] expand correctly}$ $= a^2 - 12a^2 - ab + b^2$ $= -11a^2 - ab + b^2 \text{ --- [A1]}$
2	(a) $x^4 - 16y^4$ $= (x^2 - 4y^2)(x^2 + 4y^2) \text{ --- [M1]}$ $= (x - 2y)(x + 2y)(x^2 + 4y^2) \text{ --- [A1]}$ (b) $\frac{x^4 - 16y^4}{2x^2 - 3xy + 2y^2}$ $= \frac{(x - 2y)(x + 2y)(x^2 + 4y^2)}{(2x + y)(x - 2y)} \text{ --- [M1] factorise } 2x^2 - 3xy + 2y^2$ $= \frac{(x + 2y)(x^2 + 4y^2)}{2x + y} \text{ --- [A1]}$
3	(a) $T_5 = 4^2 + 24 = 40$ (b) As each term is made up of either squared of odd terms and an odd term, or squared of even terms and an even term, each term will always be even. [M1] (c) $T_n = (n - 1)^2 + 4 + (n - 1)(5)$ $= n^2 - 2n + 1 + 4 + 5n - 5$ $= n^2 + 3n$ [M1] for any number sequence [M1] Simplify to show (d) $T_{n+1} - T_n = (n + 1)^2 + 3(n + 1) - (n^2 + 3n)$ $= n^2 + 2n + 1 + 3n + 3 - n^2 - 3n$ $= 2n + 4$ [M1] for expanding and simplifying [A1] (e) $2n + 4 = 5$ When $n = \frac{1}{2}$ Since n is not a positive integer, the difference cannot be 5.

4	$\frac{6a^2}{2a^2-8} - \frac{3a}{a+2}$ $= \frac{6a^2}{2(a-2)(a+2)} - \frac{3a}{a+2}$ $= \frac{3a^2}{(a-2)(a+2)} - \frac{3a(a-2)}{(a-2)(a+2)}$ $= \frac{3a^2 - 3a^2 + 6a}{(a-2)(a+2)}$ $= \frac{6a}{(a-2)(a+2)} \text{ or } \frac{6a}{a^2-4}$												
5	(a) $(p-5)(2p^2-p)$ $= 2p^3 - p^2 - 10p^2 + 5p$ $= 2p^3 - 11p^2 + 5p$ (b) $\frac{10-13x-3x^2}{3ax+3bx-2a-2b}$ $= \frac{(5+x)(2-3x)}{3x(a+b)-2(a+b)}$ $= \frac{(5+x)(2-3x)}{(a+b)(3x-2)}$ $= -\frac{5+x}{a+b}$												
6	<table border="1" style="width: 100%; border-collapse: collapse;"> <tr> <td style="width: 10%; padding: 5px;">ai</td><td style="padding: 5px;">11 white dots; 10 black dots</td></tr> <tr> <td style="width: 10%; padding: 5px;">aii</td><td style="padding: 5px;">$2k+1$</td></tr> <tr> <td style="width: 10%; padding: 5px;">aiii</td><td style="padding: 5px;">Total no. of dots = $\frac{1}{2}k(k-1) + (2k+1)$</td></tr> <tr> <td style="width: 10%; padding: 5px;">(a)</td><td style="padding: 5px;"> $= \frac{1}{2}k^2 - \frac{1}{2}k + 2k + 1$ $= \frac{1}{2}k^2 + \frac{3}{2}k + 1$ </td></tr> <tr> <td style="width: 10%; padding: 5px;">aiii</td><td style="padding: 5px;">$\frac{1}{2}k^2 + \frac{3}{2}k + 1 = 351$</td></tr> <tr> <td style="width: 10%; padding: 5px;">(b)</td><td style="padding: 5px;"> $k^2 + 3k - 700 = 0$ $(k+28)(k-25) = 0$ $k = -28 \text{ (N.A.) or } k = 25$ $\therefore$ Diagram 25 </td></tr> </table>	ai	11 white dots; 10 black dots	aii	$2k+1$	aiii	Total no. of dots = $\frac{1}{2}k(k-1) + (2k+1)$	(a)	$= \frac{1}{2}k^2 - \frac{1}{2}k + 2k + 1$ $= \frac{1}{2}k^2 + \frac{3}{2}k + 1$	aiii	$\frac{1}{2}k^2 + \frac{3}{2}k + 1 = 351$	(b)	$k^2 + 3k - 700 = 0$ $(k+28)(k-25) = 0$ $k = -28 \text{ (N.A.) or } k = 25$ $\therefore$ Diagram 25
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(b)	$k^2 + 3k - 700 = 0$ $(k+28)(k-25) = 0$ $k = -28 \text{ (N.A.) or } k = 25$ $\therefore$ Diagram 25												

	bi <u>Method 1</u> $T_n = S_n - S_{n-1}$ $= 3n^2 + 8n - [3(n-1)^2 + 8(n-1)]$ $= 3n^2 + 8n - [3n^2 - 6n + 3 + 8n - 8]$ $= 3n^2 + 8n - 3n^2 - 2n + 5$ $\therefore T_n = 6n + 5 \quad (\text{shown})$ bi <u>Method 2</u> $T_1 = 3(1)^2 + 8(1) = 11$ $T_2 = 3(2)^2 + 8(2) - 11 = 17$ $T_3 = 3(3)^2 + 8(3) - 17 - 11 = 23$ $T_4 = 3(4)^2 + 8(4) - 23 - 17 - 11 = 29$ $\therefore T_n = 6(n-1) + 11$ $= 6n - 6 + 11$ $= 6n + 5 \quad (\text{shown})$ bii $T_p = 125$ $6p + 5 = 125$ $p = \frac{125 - 5}{6} = 20$ $\therefore T_{p-5} = 6(15) + 5 = 95$
7	(a) $= 16n^2 - [(4n)^2 - a^2]$ $= a^2$ (b) $= (75)^2 - [(75+3)(75-3)] \quad \text{M1}$ $= 3^2$ $= 9$
8	(bi) $4x^2 + 5x - 6 = (4x - 3)(x + 2)$ (bii) $4(2t + 2)^2 + 5(2t + 2) - 6$ $= [4(2t + 2) - 3][(2t + 2) + 2]$ $= (8t + 5)(2t + 4)$ $= 2(8t + 5)(t + 2)$
9	$\frac{4(x-2)}{x^2-4} - \frac{2(3x-1)}{3x^2+5x-2}$ $= \frac{4(x-2)}{4(x-2)} - \frac{2(3x-1)}{2(3x-1)}$ $= \frac{(x-2)(x+2)}{(x-2)(x+2)} - \frac{(3x-1)(x+2)}{(3x-1)(x+2)}$ $= \frac{4}{x+2} - \frac{2}{x+2}$ $= \frac{2}{x+2}$

10	(a) $T_1 = 17$ $T_2 = x = 17 + 1(-2) = 15$ $T_3 = y = 17 + 2(-2) = 13$ $T_4 = z = 17 + 3(-2) = 11$ $T_5 = 23 = 17 + (5 - 1)(-2) = 9$ <i>Alternative method</i> Common difference = 2 $x = 17 - 2 = 15$ $y = 17 - 4 = 13$ $z = 17 - 6 = 11$ (b) $T_n = 17 + (-2)(n - 1)$ $T_n = 19 - 2n$ (c) $19 - 2n = -317$ $n = 168$ Since n is a positive integer, -317 is the 168th term of the sequence. OR Since n is a positive integer, -317 is a term of the sequence.		
11	a $\frac{9 - 25y^2}{5xy - 10y - 3x + 6}$ $= \frac{(3 - 5y)(3 + 5y)}{5y(x - 2) - 3(x - 2)}$ $= \frac{3 - 5y}{x - 2} \cdot \frac{3 + 5y}{5y - 3}$ $= \frac{3 + 5y}{2 - x}$ b $\frac{3}{2 - x} - \frac{1}{x - 2} + 5$ $= \frac{3}{2 - x} + \frac{1}{2 - x} + 5$ $= \frac{4}{2 - x} + \frac{5(2 - x)}{2 - x}$ $= \frac{14 - 5x}{2 - x}$ or $\frac{3}{2 - x} - \frac{1}{x - 2} + 5$ $= \frac{3(x - 2) - (2 - x) + 5(2 - x)(x - 2)}{(2 - x)(x - 2)}$ $= \frac{3x - 6 - 2 + x + 5(2x - 4 - x^2 + 2x)}{(2 - x)(x - 2)}$ $= \frac{4x - 8 + 20x - 20 - 5x^2}{(2 - x)(x - 2)}$ $= \frac{24x - 28 - 5x^2}{(2 - x)(x - 2)}$ $= \frac{14 - 5x}{2 - x} \cdot \frac{x - 2}{x - 2}$ $= \frac{14 - 5x}{2 - x}$ c $\text{Value} = x \left(1 - \frac{3}{100} \right)^{24}$ $= x \cdot 0.97^{24} \quad \text{or} \quad = x \left(\frac{97}{100} \right)^{24}$ $= \$0.481x \text{ (3sf)}$		

